

3	(a)	The internal energy of a system is the sum of a random distribution of kinetic and potential energies associated with the molecules of the system.		B1
	(b)	(i)	$Q = mc\Delta\theta$ $= (1.28 \times 0.600)(1000)(390 - 275)$ C1 $= 88.3 \text{ kJ}$ A1	
		(ii)	$pV = nRT$ or NkT $V \propto T$ $\frac{V_1}{V_2} = \frac{T_1}{T_2}$ $\frac{0.600}{V_2} = \frac{275}{390}$ $V_2 = 0.850 \text{ m}^3$ C1 $p\Delta V = (1.03 \times 10^5)(0.850 - 0.600)$ M1 $= 25.8 \text{ kJ}$ A0	
		(iii)	$\Delta U = W + Q$ $= -25.8 + 88.3$ M1 (allow ecf from (i)) $= 62.5 \text{ kJ}$ A1 (Cannot use $E_K = 3/2NkT$ since air in this question is not monatomic)	
	(c)	Air compressed: <u>work done on air (W) is positive</u> ; Process is sudden: <u>no heat flow (or Q is zero)</u> Therefore, <u>by the First Law of Thermodynamics, internal energy increases.</u> Since ideal gas, internal energy is proportional to temperature, hence <u>temperature increases.</u>		B1 M1 A1

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4	(a)	To the left or from B to A	R1
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